Architecture - MultiNodeRoutingEngine - Spec

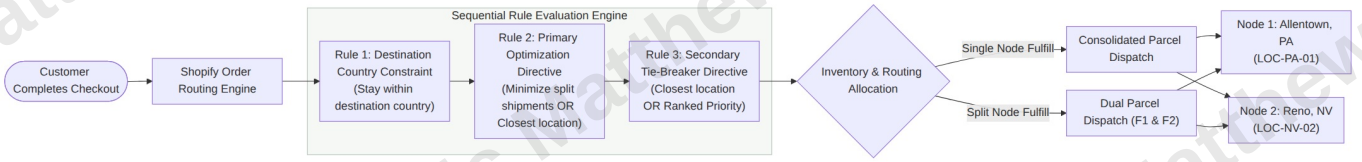
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Document ID:     Architecture - MultiNodeRoutingEngine - Spec
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        Architecture (Systems Engineering & Routing Specification)
Purpose:         MultiNodeRoutingEngine
Tier / Level:    Spec (Production Configuration & Rule Governance)
Status:          Locked 🔒 (Master Baseline)
Target Audience: TechOps Engineers, Logistics Leads, Systems Architects
Ecosystem Scope: Shopify Native Order Routing Engine, Shopify Inventory API, ShipBob / ShipBob WMS
Business Unit:   Synapse Botanicals (Multi-Node DTC Fulfillment)
```

1. Executive Summary & Network Topology

Synapse Botanicals operates a distributed direct-to-consumer (DTC) fulfillment network across the contiguous United States. Orders are fulfilled dynamically using **Shopify's Native Order Routing Engine** (`Settings` → `Shipping and delivery` → `Order routing`).

This specification establishes:

1. **Network Node Architecture:** Distribution roles and SKU stocking profiles for East Coast and West Coast nodes.
2. **Algorithmic Evaluation Hierarchy:** The strict execution order of Shopify's routing rule stack.
3. **The Cross-Country Routing Trap:** Technical analysis of how rule priority alters parcel dispatch decisions.
4. **Safety Stock Buffer Thresholds:** Mathematical buffers preventing stockouts and unintentional split shipments.
5. **Operational Location Reassignment Protocols:** Procedures for manually overriding fulfillment nodes.

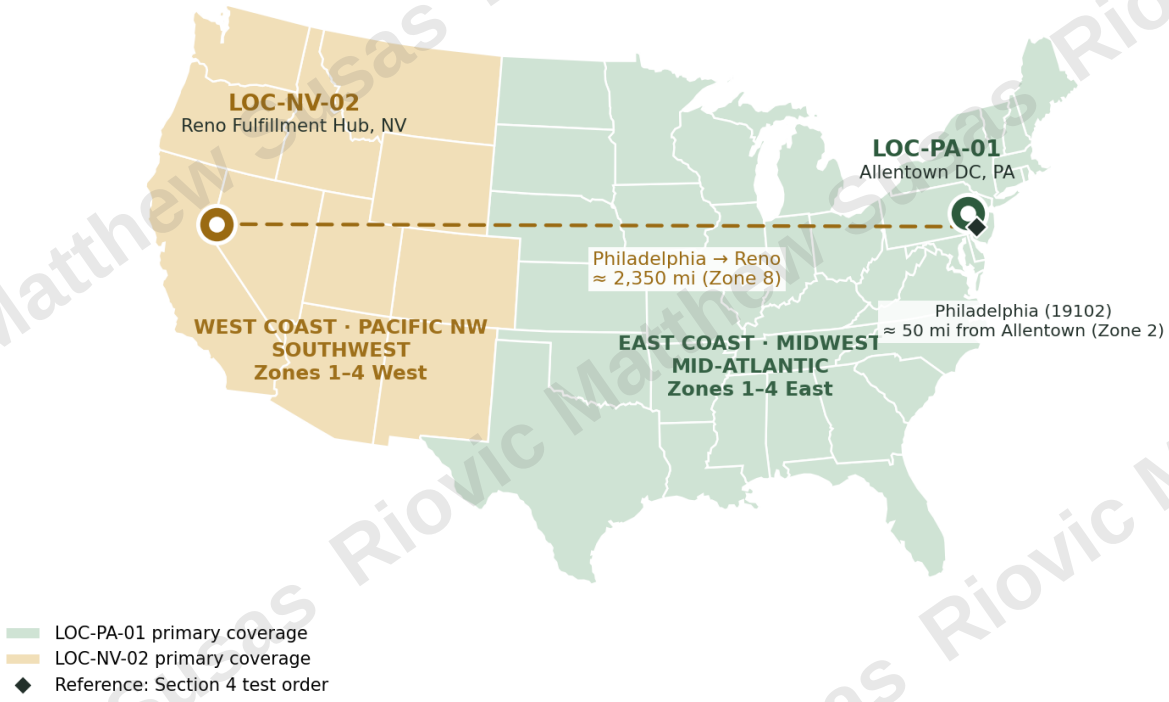


2. Fulfillment Node Profiles & SKU Stocking Topology

To optimize delivery speed and reduce carrier shipping zones, inventory is partitioned across two strategic distribution centers:

2.1 Node Profiles

Node Identifier	Facility Name	Geographic Focus	Transit Zone Coverage	Primary Facility Role
`LOC-PA-01`	Allentown Distribution Center, PA	East Coast, Midwest, Mid-Atlantic	Zones 1–4 East	High-velocity consumable hub (Stick Packs & Starter Bundles)
`LOC-NV-02`	Reno Fulfillment Hub, NV	West Coast, Pacific NW, Southwest	Zones 1–4 West	Specialty tincture hub + hardware accessory stock + West consumable hub



Illustrative coverage based on the Geographic Focus column; actual routing is determined by Rule 3 distance, not state lines.

Figure 2.1: Node coverage and proximity – LOC-PA-01 (Allentown, PA) and LOC-NV-02 (Reno, NV)

2.2 SKU Stocking & Availability Matrix

Not all SKUs are dual-stocked across both warehouses due to manufacturing batch sizes and storage requirements:

Product Title	SKU Identifier	Allentown, PA (`LOC-PA-01`)	Reno, NV (`LOC-NV-02`)	Stocking Strategy
Focus Stick Packs (Matcha Yuzu)	`SYN-FSTK-MY`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked (Universal velocity)
Focus Stick Packs (Blood Orange)	`SYN-FSTK-BO`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked (Universal velocity)
Focus Stick Packs (Wild Blueberry)	`SYN-FSTK-WB`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked (Universal velocity)
Focus Stick Packs (Crisp Green Apple)	`SYN-FSTK-GA`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked (Universal velocity)
Adaptogen Tincture (AM Focus)	`SYN-TNCT-AM`	*— (Not Stocked)*	**Exclusive Hub (100%)**	Single-Node Specialized Hub
Adaptogen Tincture (PM Restore)	`SYN-TNCT-PM`	*— (Not Stocked)*	**Exclusive Hub (100%)**	Single-Node Specialized Hub
The Daily Protocol Starter Bundle	`SYN-BNDL-MAO / SYN-BNDL-BAB`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked Kitted Unit
Precision Insulated Tumbler	`SYN-TMBL-MO / BW / SG`	**Primary Stock**	**Secondary Stock**	Dual-Stocked Hardware
Micro-Foam Electric Whisk	`SYN-ACC-WHSK`	*— (Not Stocked)*	**Exclusive Hub (100%)**	Single-Node Accessory Stock

3. Algorithmic Routing Rule Hierarchy

Shopify evaluates order routing rules **sequentially from top to bottom**. Each rule acts as a filter or a sort function on the list of eligible fulfillment locations.

3.1 Production Rule Stack Configuration

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│ SHOPIFY ORDER ROUTING RULE STACK (PRODUCTION CONFIGURATION)                  │
├──────┬──────────────────────────────────┬──────────────┬─────────────────────┤
│ Rank │ Rule Type                        │ Action Mode  │ Algorithmic Purpose │
├──────┼──────────────────────────────────┼──────────────┼─────────────────────┤
│ 1    │ Stay within destination country  │ Hard Filter  │ Eliminate foreign locations  │
│ 2    │ Minimize split shipments         │ Optimization │ Eliminate multi-box splits   │
│ 3    │ Fulfill from closest location    │ Sort / Tie   │ Minimize carrier transit zone│
│ 4    │ Ranked locations (Fallback)      │ Default Tie  │ Default: LOC-PA-01 > NV-02   │
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3.2 Detailed Execution Mechanics

Rule 1: Stay within destination country (Hard Filter)

- **Mechanics:** Compares the order's `shipping_address.country_code` against location country codes.
- **Result:** Any location outside the customer's sovereign border is permanently removed from the candidate pool for that order.

Rule 2: Minimize split shipments (Primary Optimization)

- **Mechanics:** Evaluates all possible combinations of candidate locations that can fulfill the entire line item set:
  - **Option A (Single Location Candidate):** If any single warehouse contains 100% of the ordered items in Available-to-Sell (ATS) stock, Shopify **discards all multi-location combinations**.
  - **Option B (Multi-Location Fallback):** If no single warehouse has all items, the engine calculates the minimum number of locations required (e.g., 2 locations instead of 3).

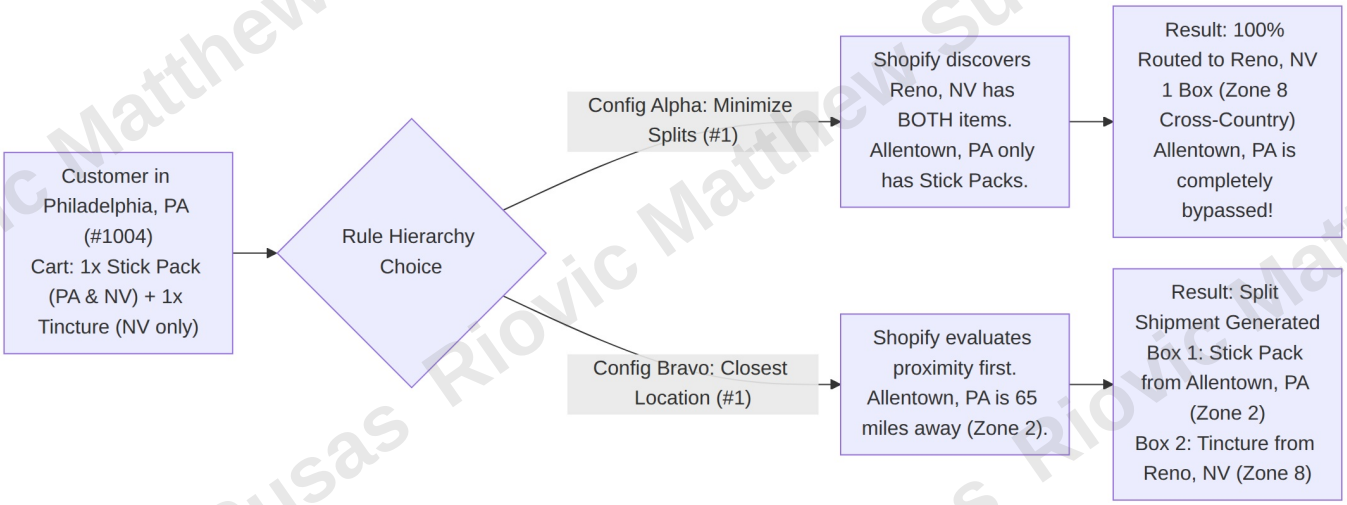
Rule 3: Fulfill from closest location (Distance Sort & Tie-Breaker)

- **Mechanics:** Calculates straight-line geodesic distance between the customer's delivery ZIP code and the geocoded address of eligible candidate locations.
- **Tie-Breaker:** When two or more single-location candidates can fulfill the entire order, the engine selects the facility closest to the recipient.

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4. The "Hidden Gap" Exposed: Split-Averse vs. Proximity Routing Trap

The most common operational failure in multi-warehouse e-commerce is the **Cross-Country Routing Trap**. When operators do not understand how Rule 2 interacts with Rule 3, they mistakenly believe the warehouse integration is defective.



4.1 Comparative Scenario Analysis

Test Order Parameters:

- **Customer Location:** Philadelphia, PA (`19102`)
- **Basket Items:**
  - 1x `Focus Stick Packs (30-Pack)` – Dual-stocked at `LOC-PA-01` and `LOC-NV-02`
  - 1x `Adaptogen Tincture (60ml)` – Stocked **exclusively** at `LOC-NV-02`

System Metric	Configuration Alpha (Split-Averse: Production)	Configuration Bravo (Proximity-Averse: Split Allowed)
Rule Order	1. Minimize split shipments 2. Closest location	1. Fulfill from closest location 2. Minimize split shipments
Assigned Warehouse(s)	`LOC-NV-02` **(Reno, NV) exclusively**	`LOC-PA-01` **(PA) &** `LOC-NV-02` **(NV) [SPLIT]**
Packages Dispatched	**1 Package** (Both items consolidated)	**2 Packages** (Dual split packages)
Carrier Zone(s)	Zone 8 (Cross-country: Reno → Philly)	Package 1: Zone 2 (Allentown → Philly) Package 2: Zone 8 (Reno → Philly)
Outbound Postage Cost	**$11.45** (Consolidated 1.2 lb box)	**$17.20** ($6.40 Zone 2 + $10.80 Zone 8)
Transit Time (SLA)	4–5 Business Days (Ground)	Stick Pack: 1–2 Days | Tincture: 4–5 Days
Customer Experience	1 Box, 1 Tracking Number	2 Boxes arriving on different days
Operator Perception	"*Why did PA get skipped? Is PA broken?*"	"*Why did this split? We lost $5.75 on postage!*"

> **IMPORTANT THE PRODUCTION DIRECTIVE**
> Synapse Botanicals maintains **Configuration Alpha** as the production baseline. Ingestible botanical drink mixes and tinctures are routinely consumed together as a morning ritual. Consolidating into a single package saves **$5.75 per order in baseline carrier overhead** and eliminates customer confusion caused by staggered delivery dates.

> **NOTE Empirical Validation Reference:**
> For an end-to-end audit trail of how this exact split-routing scenario was processed, intercepted, and reconciled with live customer exceptions, review **Audit - OrderStateSimulation - RunSheet (Domain 1.3)** – **Section 3**.

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5. Inventory Buffers & Safety Stock Governance

To prevent stockouts from high-velocity sales spikes and avoid race conditions between Shopify and 3PL WMS synchronization cycles, safety stock buffers must be established.

5.1 Buffer Mathematics & Available-to-Sell (ATS) Logic

Shopify calculates fulfillable inventory using the following formula:

$$\text{ATS (Available-to-Sell)} = \text{On-Hand} - \text{Committed} - \text{Safety Buffer}$$

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│ INVENTORY SAFETY BUFFER SPECIFICATION                                    │
├──────────────────────────┬──────────────┬──────────────┬──────────────┬──────────────┤
│ Product Category         │ Velocity Tier│ PA Buffer    │ NV Buffer    │ Rationale    │
├──────────────────────────┼──────────────┼──────────────┼──────────────┼──────────────┤
│ Focus Stick Packs (All SKUs) │ High (Tier 1)│ 15 Units     │ 10 Units     │ Rapid spikes │
│ Adaptogen Tinctures      │ Med  (Tier 2)│ –            │ 10 Units     │ Single-hub   │
│ Starter Protocol Bundles │ Med  (Tier 2)│ 10 Units     │ 5 Units      │ High AOV kit │
│ Tumblers & Whisks        │ Low  (Tier 3)│ 5 Units      │ 5 Units      │ Stable drain │
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5.2 Dynamic Out-of-Stock Fallback Cascade

When a distribution node's ATS reaches ≤ 0 (due to on-hand inventory hitting the safety buffer):

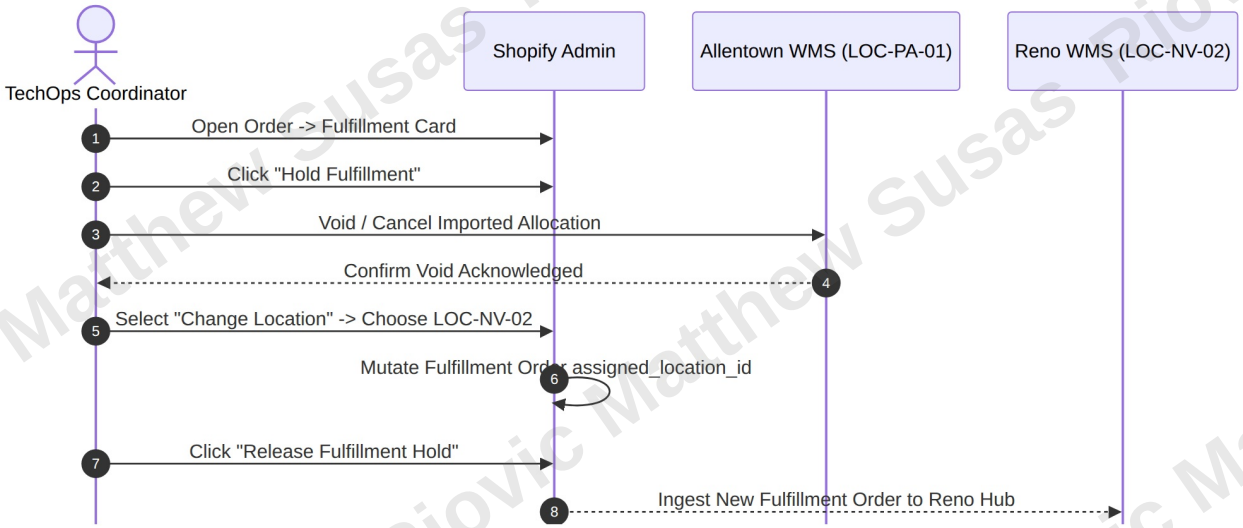
1. **Automated De-allocation:** The native routing engine automatically disqualifies that location from candidate pools for that SKU.
2. **Cascade to Secondary Node:** Fulfillments dynamically transfer to the secondary node (`LOC-NV-02` for Stick Packs).
3. **Low-Stock Notification Trigger:** If total network ATS falls below 25 units, Shopify Flow generates an automated Slack notification to `#supply-chain-alerts`.

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6. Operational Location Reassignment Protocol (Manual Overrides)

When physical warehouse constraints (e.g., weather delays, carrier trailer shortages, physical inventory audits) require an operational override, TechOps coordinators must execute a manual location reassignment.

Riovic Matthew Susas



6.1 Step-by-Step Location Override Procedure

1. **Place Fulfillment Hold:** Never change locations on an active, unheld fulfillment order. Apply hold code: `Inventory issue`.
2. **Void Original 3PL Record:** Access the source warehouse portal (e.g., ShipBob Allentown). Search Order ID → Click **Cancel / Void Order**. Confirm the shipment is not in a wave pick.
3. **Execute Location Mutation in Shopify Admin:**
   - Navigate to the target fulfillment card.
   - Click the **three horizontal dots (...)** → Select **Change location**.
   - In the location dropdown, select the target facility (e.g., `Reno Fulfillment Hub`).
   - Verify line items and quantities transfer completely.
   - Click **Save**.
4. **Release Hold:** Release the Shopify fulfillment hold. Confirm the order transmits to the destination 3PL queue via the `fulfillment_orders/scheduled` topic.
5. **Tag Order:** Append tracking tag: `OPS: Rerouted [Old_Loc] to [New_Loc]`.

> **NOTE System Governance & Safety Dependency:**
> Location reassignments must strictly follow the native hold protocol in **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)** – **Section 2**, and are strictly prohibited once an order crosses the Point of No Return defined in **Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)** – **Section 3**.

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7. Change Management & Verification Checklist

Modifications to the Shopify Order Routing rule stack impact 100% of incoming customer orders instantly. Any proposed changes to the rule order must undergo the following release gate:

- [ ] **Off-Peak Execution:** Routing rule adjustments must be deployed Tuesday–Thursday between 10:00 PM and 11:30 PM EST.
- [ ] **Pre-Flight Test Suite:** Run 4 synthetic draft orders representing:
  1. East Coast Single-Node basket (PA only)
  2. West Coast Single-Node basket (NV only)
  3. East Coast Mixed basket (PA & NV SKUs)
  4. Out-of-Stock Fallback basket (PA inventory depleted)
- [ ] **3PL Ingestion Validation:** Confirm test order fulfillment orders are ingested into the correct warehouse WMS tenant within 180 seconds.
- [ ] **Post-Deploy Audit:** Review the first 25 live production orders following deployment to ensure fulfillment card locations match the expected model.